

Dawson Butler

Charlotte, NC | Knoxville, TN | 980 287 6058 | dawsgbutler@gmail.com | www.linkedin.com/in/dawsonbutler

EDUCATION

Bachelor of Science in Mechanical Engineering | *University of Tennessee, Knoxville*
Minor in Aerospace Engineering | GPA 3.27 | Dean's List 2025

Expected May 2028

TECHNICAL SKILLS

CAD & Design	Autodesk Revit, Autodesk Inventor, AutoCAD, Parametric Modeling, Revit Family Creation, Mechanical Drafting
Construction & VDC	Bluebeam Revu, BIM/VDC Coordination, Autodesk Construction Cloud (ACC), Evolve, Shop & Installation Drawings, Plan Review, Microsoft Excel
Manufacturing	Robotic Assembly, Electromechanical Systems, Wiring & Troubleshooting, 3D Printing, Prototyping, Mechanical Fabrication, Hand & Power Tools
Programming	API Integration, Basic Python/Matlab, Automation, Backtesting, Data Analysis

EXPERIENCE

College Project Engineer Intern, Virtual Design and Construction | *Gaylor Electric*

May 2026 to Present

Charlotte, NC

- Supported electrical construction and VDC workflows for large-scale data center projects.
- Built and revised Revit model content from AutoCAD files, project prints, and existing plan information.
- Assisted with Revit family creation and parametric modeling for modular skid and mounting plate components.
- Helped identify drawing conflicts, constructability questions, installation details, and coordination issues before field installation.
- Supported shop drawing and installation drawing workflows by organizing model information and checking dimensions against project prints.

Robotics Engineering Technician | *LUCID Bots*

May 2025 to Present | Previous Summer 2023

Charlotte, NC

- Built industrial robotic cleaning systems from mechanical assembly through wiring, testing, and final quality inspection.
- Troubleshoot fit issues, wiring problems, assembly errors, damaged components, and quality control concerns during robot production.
- Communicated recurring production issues and process improvement opportunities to engineers and team leads.
- Helped move production from a backlog of roughly 50 robots toward on-time delivery by supporting full system builds & production flow improvements.
- Rebuilt and tuned a 3D printer used for prototyping and internal parts, then produced printed components for shop and development use.

LEADERSHIP

Director of Standards and Construction Project Lead | *Delta*

2025 to Present

Tau Delta Fraternity

Knoxville, TN

- Serve as Director of Standards while leading chapter house construction, maintenance, and improvement projects.
- Helped manage an \$8,000 Homecoming budget while coordinating materials, labor, schedules, and build responsibilities.
- Led hands-on projects involving stages, full wooden bars, basement improvements, repairs, and house upgrades.
- Designed and fabricated custom wooden sorority letters and event displays, managing design, material planning, fabrication, and delivery.

Certifications

- Autodesk Certified User: Inventor *2024*
- Autodesk Certified User: AutoCAD *2023*

PROJECTS

CAD & 3D Printing Projects | *Autodesk Inventor, Revit, AutoCAD, 3D Printing, Slicing, Optimization*

- Designed and fabricated custom parts, shop fixtures, home improvement components, and functional prototypes using CAD and 3D printing.
- Optimized parts for strength, printability, and ease of assembly through multiple design iterations.
- Produced replacement parts and custom solutions for personal and engineering projects.

AI Trading Decision System | *Python, Alpaca API, Automation*

- Developed a Python-based automated trading system that scans stocks, evaluates trade opportunities, executes paper trades, and tracks performance.
- Built decision logic using technical indicators, market conditions, and risk management.
- Implemented automated reporting, back testing, replay testing, and order management.

Construction Safety Analytics Platform | *QR Codes, Dashboard Design, Process Improvement*

- Designed a QR code-based near miss reporting system to identify jobsite hazards before recordable incidents occur.
- Developed a dashboard concept for tracking hazard trends, response times, and corrective actions.